Andreas Makrides

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Education

Imperial College London — *London, UK*
PhD in Machine Learning & Optimisation for Electricity Markets — Oct 2025 – Present

- Designing new **pricing mechanisms**, **machine-learning methods for large-scale optimisation** and **trading strategies** for electricity markets with high shares of renewables and battery storage.
- **Won** the 2025 EPRI AI-ccelerating Unit Commitment Competition with an **optimisation-aware machine learning** method, achieving a mean optimality gap below **0.5%** and an average speed-up of more than **20**×.
- Elective coursework: Probability for Statistics, Statistical Methods for Finance.

Imperial College London — *London, UK*
MSc in Sustainable Energy Futures — Sep 2024 – Sep 2025

- Grade: **Distinction**.
- Thesis: *Incomplete Markets for Risk and Storage Duration in Future Electricity Markets* (75%).
- Selected modules: Energy Economics & Policy (81.17%), Methods for the Analysis of Energy Systems – Scientific Computing (79.13%), Data Science & Digitalisation in the Energy Sector (70%).
- **Outstanding Team Member Award**, Sustainable Energy Futures Conference 2025.

University of Cyprus — *Nicosia, Cyprus*
BSc in Electrical Engineering — Sep 2020 – May 2024

- Grade: **9.89/10**, Valedictorian (Class of 2024), **President of the Republic of Cyprus Award**.
- Thesis: *Advanced Real-Time Simulation Modelling and State Estimation of the Smart Energy Campus of the University of Cyprus* (10/10).
- Selected modules (all 10/10): Linear Algebra, Differential Equations, Numerical Analysis, Probability, Statistics, Signals & Systems, Control, Power Engineering.
- **Junior Achievement (JA) StartUp**: 1st place in the National Innovation Competition (Cyprus) and represented Cyprus at GEN-E 2024, Europe's largest entrepreneurship festival.

Experience

Systematic Momentum Research — *London, UK*
Personal Project — Oct 2026 – Present

- Building a tested **Python** backtesting engine (no look-ahead bias) for **time-series** and **cross-sectional momentum** on ETFs and S&P 500 stocks, using survivorship-free CRSP data.
- Implementing **EWMA volatility targeting** and **Ledoit–Wolf mean–variance optimisation**, benchmarking against 1/N and minimum-variance portfolios.
- Controlling **overfitting** with a 5-year holdout, walk-forward testing and deflated Sharpe ratios.
- **Paper trading** monthly rebalances on Interactive Brokers and reconciling fills with the backtest.

PHAETHON Centre of Excellence — *Nicosia, Cyprus*
Research Scientist — Jan 2024 – Aug 2024

- Built **simulation and control software** for microgrids, with a **real-time** interface for monitoring and analysing live data from operational energy systems.

FOSS Research Centre for Sustainable Energy — *Nicosia, Cyprus*
Research Scientist — Jun 2021 – Dec 2023

- Developed **state-estimation** methods that infer unobserved system states from noisy, real-time measurements in microgrids and smart energy systems.
- Built **predictive models** and **forecasts** to support the operation of these systems.

Texas A&M University — *College Station, Texas, USA*
Research Intern — Jun 2023 – Jul 2023

- Quantified the impact of **crypto-mining** load on **electricity market prices** and spatial price resolution.
- Forecast Bitcoin usage with **time-series** and **machine-learning** models.

Cyprus National Guard — *Nicosia, Cyprus*
Soldier — Jul 2019 – Sep 2020

- Completed 14 months of compulsory military service.

Publications

C = Conference, S = In Submission

[S.1] **A. C. Makrides**, A. Suski and E. Spyrou, "**Quantifying the Impact of Missing Risk Markets on Investment in Decarbonized Power Systems with Long-Duration Energy Storage**," submitted to *Sustainable Energy, Grids and Networks*, 2026.

[S.2] B. Fritz, **A. C. Makrides**, M. Fetanat and P. Pinson, "**Learning to Fix: Optimisation-Aware Machine Learning for Accelerated Unit Commitment**," submitted to *IEEE Transactions on Power Systems*, 2026.

[C.1] **A. C. Makrides** et al., "**Predicting Energy Market Imbalance Prices with Random Forest: A Rolling Window Approach**," *2024 3rd International Conference on Energy Transition in the Mediterranean Area (SyNERGY MED)*, Limassol, Cyprus, 2024.

Skills

Technical Skills

- **Programming:** Python, Julia, MATLAB (Advanced), R and C (Intermediate)
- **Data Science & Machine Learning:** NumPy, Pandas, Scikit-learn, PyTorch
- **Optimisation:** convex and non-convex (MILP, MISOCP, bilinear), bilevel and equilibrium models (complementarity), optimisation under uncertainty (chance-constrained, CVaR risk-averse), decomposition methods (Dantzig–Wolfe, ADMM)
- **Statistics & Forecasting:** time-series forecasting, state estimation, statistical modelling, backtesting
- **ECDL Certificate (7 modules):** Computer Essentials, Online Essentials, Word Processing, Spreadsheets, Presentation, IT Security, Online Collaboration

Additional Skills

- **Languages:** Greek (Native), English (Fluent)

Extracurricular Activities & Interests

- **Activities:** Radio Amateur, Deputy Scout Leader (273rd Air Scouts) and recipient of the Certified Scout of Democracy Award from the 7th President of the Republic of Cyprus, Nicos Anastasiades
- **Sports:** running (London Marathon 2027, fundraising for the MND Association), padel, football
- **Interests:** financial markets, quantitative trading strategies, investing